$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1-\lambda & 1 \\ 1 & -1-\lambda \end{bmatrix}$$

$\det(A - \lambda I) \Rightarrow (1-\lambda)(1+\lambda) + 1 = 0$

$(1-\lambda^2) + 1$

$\lambda^2 = 2$

$\lambda_1 = \sqrt{2}$

$\lambda_2 = -\sqrt{2}$

eigen values.

$\det(A^{13} - \lambda I) = 0$:-

$\hookrightarrow (64-\lambda)(64+\lambda) + (64)^2 = 0$

$A^2 = 2I$

$A^{13} = 64A$

$2(64)^2 = \lambda^2$

$\lambda = +64\sqrt{2}$; $\lambda = -64\sqrt{2}$